

Data Collection and Preprocessing Phase

Date	26 June 2026
Team ID	Individual
Project Title	Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits
Maximum Marks	3 Marks

Data Quality Report Template

This Data Quality Report summarizes the quality assessment of the food ordering behaviour dataset used in the project Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits. It documents identified data quality issues, their severity levels, and the resolution plans applied to ensure accurate, consistent, and reliable data for Tableau analysis and dashboard development.

Data Source	Data Quality Issue	Severity	Resolution Plan
food_ordering_behaviour_dataset.csv	No missing values detected across all 19 columns (order_id, user_id, age, city, order_time, day_type, cuisine, meal_type, restaurant_type, order_value, discount_applied, delivery_fee, time_taken_to_order, rating_given, is_repeat_order, mood, hunger_level, company, rainy_weather).	Low	No action required. Dataset is fully populated. Verified using null count check across all columns — result: 0 nulls in all 19 fields.

food_order ing_behavi or_dataset. csv	No duplicate records found. All 50,000 order_id values are unique with zero duplication across the entire dataset.	Low	No action required. Duplicate check on order_id returned 0 duplicates. Dataset integrity confirmed for all records.
food_order ing_behavi or_dataset. csv	order_value and delivery_fee stored as int64 data type, which may cause precision loss when computing ratio metrics such as Delivery Fee Percentage = $(\text{delivery_fee} / \text{order_value}) \times 100$.	Moderate	Cast order_value and delivery_fee columns to float64 before ratio calculations in Python/Tableau. Use FLOAT() conversion in Tableau calculated fields to ensure decimal precision.
food_order ing_behavi or_dataset. csv	rating_given is stored as an integer (1-5) but functions as a categorical-ordinal variable. Treating it purely as a continuous measure could produce misleading averages in certain visualizations.	Low	Load rating_given as a Discrete Dimension in Tableau for distribution charts. Use AVG(rating_given) only for trend comparisons. Add a filter to allow toggling between individual rating counts and average rating views.
food_order ing_behavi or_dataset. csv	Categorical columns (discount_applied, is_repeat_order, rainy_weather) use Yes/No string encoding instead of boolean (True/False or	Low	Create Tableau calculated fields to convert Yes/No to 1/0 for aggregation: IF [discount_applied] = "Yes" THEN 1 ELSE 0 END. Use these

	1/0), which requires string-based filtering logic in Tableau.		derived fields for rate calculations (Repeat Order Rate, Discount Usage Rate).
food_ordering_behavior_dataset.csv	No standardized composite keys exist for cross-dimensional analysis (e.g. no single column combining cuisine + meal_type or mood + hunger_level), limiting direct multi-attribute segmentation.	Low	Create derived calculated columns in Tableau: Cuisine-Meal = [cuisine] + " - " + [meal_type]; Behavioural Profile = [mood] + " - " + [hunger_level]; Session Label = [day_type] + " - " + [order_time]. Use these for heatmaps and cross-segment filters.
food_ordering_behavior_dataset.csv	Near-uniform distribution across cities (8,262-8,443 orders each) and cuisines (8,228-8,452 each) means no single dominant category exists, which may reduce visual contrast in bar charts and heatmaps.	Low	Apply relative ranking (RANK function in Tableau) rather than absolute counts for city and cuisine comparisons. Use color saturation scales and percentage-of-total calculations to highlight relative differences despite near-uniform raw counts.
food_ordering_behavior_dataset.csv	time_taken_to_order column name is ambiguous — it could refer to time taken to place the order (browsing/decision time,	Low	Rename the field alias in Tableau to "Order Placement Time (mins)" for clarity. Add a tooltip description in all relevant charts

	range 1-14 min) rather than delivery time, which may mislead stakeholders reviewing dashboards.		explaining: This field represents the time taken by the customer to place the order, not the delivery duration.
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